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The Franck-Hertz experiment on neon und mercury

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We carried out the Franck–Hertz experiment on neon and on mercury vapor. During the experiment the excitation energies of these gases were measured. For the neon $2p \to 3p$ transition, we obtained $E_1 = (18.85 \pm 3.6)\,\mathrm{eV}$, for the neon $3s \to 3p$ transition, $E_2 = (2.1 \pm 6.5)\,\mathrm{eV}$, and for the mercury $6\,^1S_0 \to 6\,^3P_1$ transition, $E = (4.72 \pm 0.74)\,\mathrm{eV}$.

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1 Introduction

Diodes are electrical switching elements with two terminals, nowadays typically made from semiconductors, which exhibit a rectifying effect on electric current. Even today, old vacuum diodes offer by far the cleanest switching behavior, as they do not exhibit leakage currents. The Franck-Hertz experiment investigates tube diodes filled with a specific gas at low pressure. To understand the effects arising from the interaction between electrons and the gas, it is essential to understand the functioning of a vacuum diode. The Franck-Hertz experiment allows conclusions to be drawn about the discrete energy levels of the gas and the collision interactions between electrons and atoms.

2 Theoretical background

2.1 Operating principle of a vacuum diode

A vacuum diode typically consists of an evacuated glass tube with two electrical contacts: a negatively charged cathode and a positively charged anode. The cathode is heated by applying a heating voltage U_H, which leads to thermionic emission of electrons from its surface due to the glow (thermionic) effect [2]. When a positive anode voltage U_A is applied, a portion of the emitted electrons is accelerated toward the anode, resulting in a measurable anode current I_A. If the polarity is reversed, no current flows, since electrons are only emitted from the cathode. Hence, the vacuum diode acts as a rectifier.

At sufficiently high anode voltages, the anode current is limited by the number of electrons emitted by the cathode. The current density is described by the Richardson equation,

$$j = CT^2 \exp\left(-\frac{W_A}{k_B T}\right),$$

where C is a material-specific constant, W_A is the work function of the cathode material, and k_B is Boltzmann's constant.

At low anode voltages, however, not all emitted electrons contribute to the current because of space charge effects near the cathode, which shield the electric field. In extreme cases, this field may vanish entirely at the cathode surface. In this regime, the current density is governed by the Schottky-Langmuir law [2],

$$j = \frac{4}{9}\varepsilon_0 \sqrt{\frac{2e}{m_e}} \frac{(U_A - U_K)^{3/2}}{d^2},$$

where ε_0 is the vacuum permittivity, e the elementary charge, m_e the electron mass, d the distance between cathode and anode, and U_K the contact potential. The contact potential arises from the difference in work functions of the anode and cathode materials and defines the threshold voltage at which the current begins to flow.

At small anode voltages, electrons may not have enough thermal energy to overcome the opposing field, and thus no current flows. As U_A increases, the current rises according to the Schottky-Langmuir law. Once the space charge is fully neutralized and nearly all emitted electrons reach the anode, the current saturates.

2.2 Theoretical backround of the Frank-Hertz experiment

According to quantum physics, atoms can only be excited to discrete energy levels. In fact, the Franck-Hertz experiment can be seen as confirmation of this principle, as it explains the characteristic current-voltage curve.

Electrons that do not have enough energy to excite the gas interact only elastically with the atoms. In the current-voltage characteristic, this leads to an initial increase in current with increasing acceleration voltage U_B, since a growing proportion of electrons gains sufficient energy to overcome the retarding voltage.

Due to their significantly smaller mass compared to atoms, electrons lose almost no energy during elastic collisions.

Once the acceleration voltage becomes large enough, the current suddenly drops, creating a local maximum. At this point, the fastest electrons have just enough energy to collide inelastically with the gas atoms, thereby exciting them. During this process, they lose the amount of energy required for excitation and can no longer overcome the retarding field.

If the acceleration voltage is increased further, even slower electrons begin to collide inelastically, and the current decreases further, forming a falling edge in the curve. At even higher voltages, electrons that have undergone a collision are re-accelerated and can again contribute to the anode current — until they gain enough energy before reaching the second grid to undergo a second inelastic collision.

At this acceleration voltage, the second major maximum occurs. The third maximum appears analogously, when the electrons have enough energy to undergo a third inelastic collision, and so on.

Similar to a vacuum diode operating in the Schottky-Langmuir regime, the anode current increases at the maxima with rising acceleration voltage. In the minima, the current does not drop to zero but also increases, as the likelihood that electrons gain enough energy to overcome the retarding field after their last inelastic collision increases with higher acceleration voltage.

3 Experimental setups

3.1 Setup for measuring the characteristics of a vacuum diode

To record the characteristic curve of the diode, the current-voltage characteristics are measured in the range from $-500\,\mathrm{V}$ to $0\,\mathrm{V}$ in steps of $100\,\mathrm{V}$ and in the positive voltage range up to $500\,\mathrm{V}$ in steps of $10\,\mathrm{V}$. Complete characteristic curves should be recorded for heating voltages $U_\mathrm{H} = 6.5\,\mathrm{V}$ and $U_\mathrm{H} = 7.5\,\mathrm{V}$.

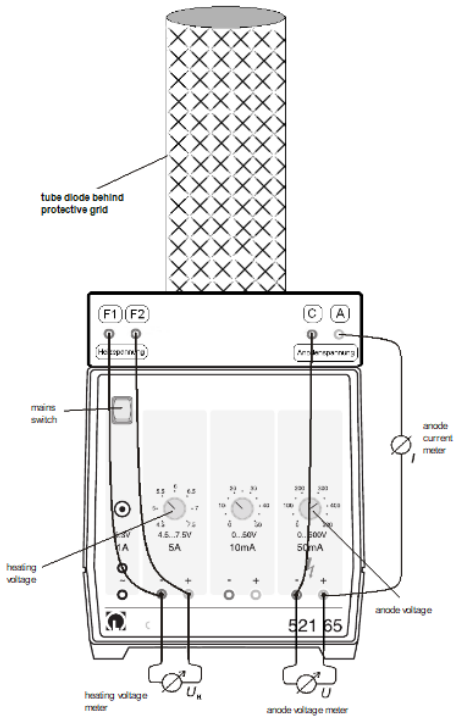

Fig. 1: Experimental setup for the vacuum diode experiment [1]

Additionally, the saturation currents at an anode voltage of $U_\mathrm{A} = 500\,\mathrm{V}$ are to be measured for heating voltages from $U_\mathrm{H} = 5.0\,\mathrm{V}$ to $U_\mathrm{H} = 7.0\,\mathrm{V}$ in steps of $0.1\,\mathrm{V}$. From the diagrams of the measured values, there can be determined which measurement points correspond to the Schottky–Langmuir space-charge region There cn also be estimated the order of magnitude of the contact potential U_K, i.e., the voltage at which the anode current vanishes.

To verify the Richardson equation, the measured saturation current values I_S at the respective heating voltages should be plotted against the temperature T. The temperature of the tungsten filament can be estimated from the heating voltage U_H using the approximate relation:

$$T(U_\mathrm{H}) = 1616\,\mathrm{K} + 97 \cdot U_\mathrm{H}\,[\mathrm{V}].$$

3.2 Setup of the Frank-Hertz experiment

The setup of the Franck-Hertz experiment consists of a gas-filled tube which contains the heated cathode and the anode, also two grids, as shown in Figure 2.

There is a controllable extraction voltage U_S applied between the first grid and the heated cathode. This voltage ensures that the shielding effects of the space charge at the cathode are suppressed for the actual experiment taking place between the grids. As a result, a portion of the electrons from the cathode is extracted and now resides between the grids, where

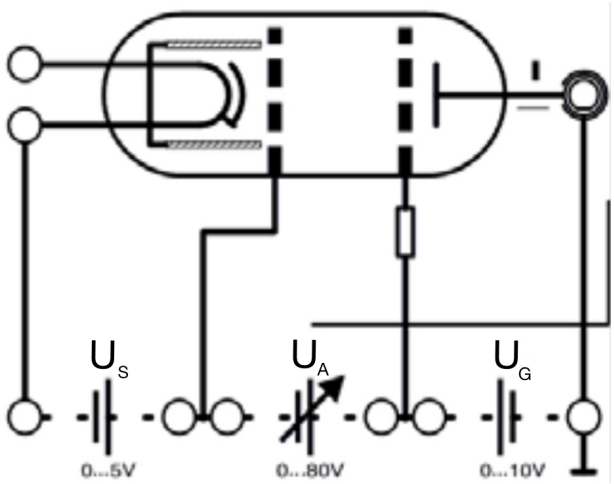

Fig. 2: Experimental setup for the Franck-Hertz experiment according to [1]

the electrons are accelerated toward the second grid by the acceleration voltage U_B.

However, only those electrons that have sufficient kinetic energy to overcome the adjustable retarding voltage U_G contribute to the anode current I_A.

Franck-Hertz tubes can be operated with different fill gases. In our experiment, neon and mercury vapor were used, and the measurement curves are shown in Figures 3 and 4. For these measurements, the retarding and extraction voltages were fixed, and the anode current was measured as a function of the acceleration voltage by recording the voltage drop across a shunt resistor. The voltage measurements were performed using a PASCO Universal Interface 550 analog-to-digital converter at a sampling rate of 50 Hz

4 Results of Franck-Hertz experiments

4.1 Franck-Hertz experiment with neon

When using neon as the fill gas, the acceleration voltage of the Franck-Hertz tube was varied from 0 V to approximately 70 V. Figure 3 shows the Franck-Hertz curve for a tube filled with neon, measured with an extraction voltage $U_1 = 0.84$ V and a retarding voltage $U_2 = 9.36$ V.

The curve closely follows the expected behavior, but in addition to the main maxima discussed above, it also exhibits secondary maxima. These can be explained by the fact that neon atoms can be excited not just to a single energy level, but to several levels between which transitions are also possible.

To quantify these observations, the distances between the maxima in the measured curve are determined. By multiplying these voltage differences with the elementary charge e, the corresponding excitation

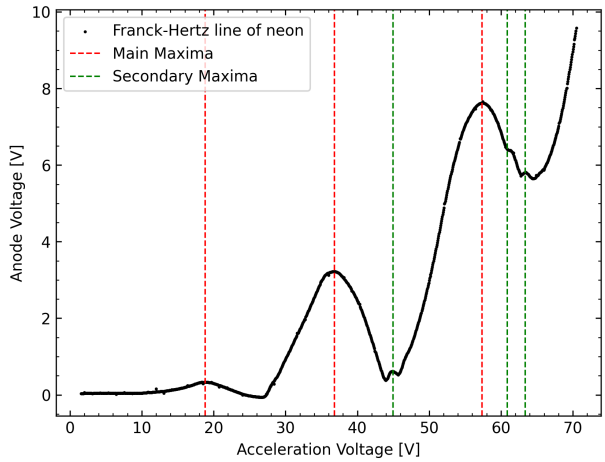

Fig. 3: Franck–Hertz curve of a tube filled with neon for an accelerating voltage $U_{av} = 0.84V$ and a retarding voltage $U_{rv} = 9.36V$. The characteristic maxima can be observed at $U_0 = (18.8 \pm 2.7)V$, $U_1 = (36.8 \pm 4.3)V$, and $U_2 = (57.3 \pm 5.7)V$. At slightly higher voltages — $U_{L1} = 20.8V$, $U_{L2} = 39.8V$, and $U_{L3} = 62.9V$ — the additional appearance of a new orange glowing plane between the grids was also observed. The highlighted secondary maxima are caused by the excitation of intermediate states and increase the spacing between the main maxima.

energies can be calculated.

The resulting energy differences are as follows:

$$\Delta E_{01} = (18.0 \pm 5.1)eV$$
$$\Delta E_{12} = (19.7 \pm 5.2)eV$$

Considering energy differences, the averaged distance between the main maxima is found to be

$$E_{1,\text{exp}} = (18.85 \pm 3.6)\,\text{eV},$$

and the distance between the last two side maxima is

$$E_{2,\text{exp}} = (2.5 \pm 6.5)\,\text{eV},$$

while the other distances between the side maxima may be explained as multiples of E_2. These can be attributed to the excitation of a neon electron from the ground state in the 2p orbital to the 3p orbital with an energy of

$$E_{1,\text{lit}} \in [18.3\,\text{eV}, 18.9\,\text{eV}],$$

or from the intermediate excited 3s orbital to the 3p orbital with an energy in the range of

$$E_{2,\text{lit}} \in [1.5\,\text{eV}, 2.3\,\text{eV}].$$

From this, we conclude that the excitation by collisions with energy E_1 from the 2p to the 3p level is

responsible for the main maxima. From the excited 3p state, the excited neon electron first decays to the 3s level under emission of yellow light, from which it either returns to the ground state or is re-excited to the 3p level by a collision with energy E_2.

This latter excitation leads to the formation of the side maxima. Since this process can occur repeatedly in quick succession, several side maxima may follow a main maximum at intervals that are multiples of E_2.

Because the characteristic curve is not ideally sharp, not all side maxima are distinguishable.

If a sufficiently large electron current is applied, the visible electromagnetic radiation resulting from the de-excitation of neon from the 3p to the 3s energy level can be observed with the eye in the form of yellow-orange glowing zones ($\lambda = 587,6\ nm$) inside the neon tube.

As the acceleration voltage increases, these luminous zones shift from the second grid toward the first grid, since the first excitation collisions occur earlier. At the same time, the zones become progressively thinner.

The acceleration voltages at which a new luminous zone just appears are:

$$U_{L1} = 20.8V$$
$$U_{L2} = 39.8V$$
$$U_{L3} = 62.9V$$

4.2 Franck-Hertz experiment with mercury vapor

Analogous to the experiment on neon we conducted one with mercury vapor at a temperature of $T = 170\,°\mathrm{C}$ as the working gas. In this setup, the acceleration voltage was varied from 0 V to approximately 30 V. The resulting curve 4 shows more main maxima better than in the case of neon, allowing a more precise quantitative analysis of the spacing between the maxima.

$$\Delta E_{01} = (4.0 \pm 1.7)eV$$
$$\Delta E_{12} = (4.6 \pm 1.7)eV$$
$$\Delta E_{23} = (4.9 \pm 1.6)eV$$
$$\Delta E_{34} = (4.9 \pm 1.6)eV$$
$$\Delta E_{45} = (5.2 \pm 1.7)eV$$

It is noticeable that the difference between neighboring maxima increases with increasing acceleration voltage. This can be attributed to the fact that an electron does not immediately collide with an atom upon reaching the excitation energy of the gas. Instead, it travels the mean free path of the gas. During this path, the electron gains energy proportional to the acceleration voltage, which is then lost in the next inelastic collision.

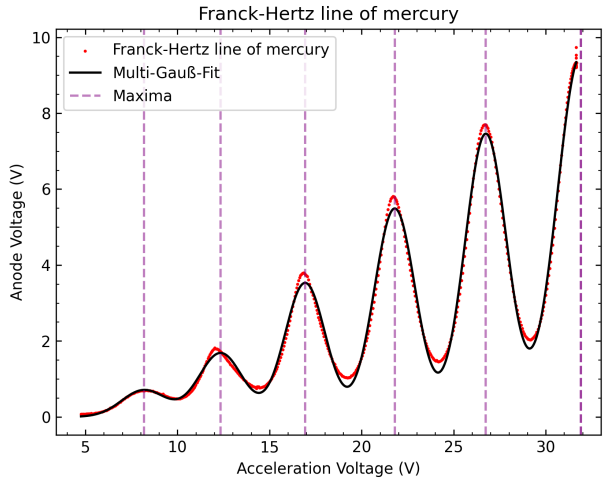

Fig. 4: Franck–Hertz curve of a tube filled with mercury vapor for an accelerating voltage $U_{av} = 4.60V$ and a retarding voltage $U_{rv} = 2.06V$. The characteristic maxima can be observed at $U_0 = (8.2 \pm 1.2)V$, $U_1 = (12.3 \pm 1.2)V$, $U_2 = (16.9 \pm 1.1)V$, $U_3 = (21.8 \pm 1.1)V$, $U_4 = (26.7 \pm 1.1)V$ and $U_2 = (31.9 \pm 1.3)V$.

Due to this effect, it is expected that the excitation energy determined from the average value of the maxima differences given above,

$$E_{\text{average}} = (4.72 \pm 0.74)\,\text{eV},$$

is slightly greater than the actual transition energy of mercury. However, because of the large error margin, this cannot be quantitatively confirmed with our result.

According to the energy level diagram of mercury, the observed transition can be identified as the transition from the $6\,^1S_0$ state to the $6\,^3P_1$ orbital, with an energy of

$$E_{\text{lit}} = 4.89\,\text{eV}\ [1].$$

Our experimentally determined value is consistent with this within the given error range.

5 Conclusion

In our experiment, the Franck–Hertz experiment was conducted using neon and mercury vapor, and the excitation energies of these gases were measured.

For the neon $2p \to 3p$ transition, we obtained

$$E_{1,\text{exp}} = (18.85 \pm 3.6)\,\text{eV},$$

and for the neon $3s \to 3p$ transition,

$$E_{2,\text{exp}} = (2.1 \pm 6.5)\,\text{eV}.$$

For the mercury $6\,^1S_0 \rightarrow 6\,^3P_1$ transition, the experimentally determined excitation energy was

$$E_{\text{exp}} = (4.72 \pm 0.74)\,\text{eV}.$$

In the case of neon, visible light emission due to atomic relaxation in the collision zones was also observed.

References

[1] Physikalisches Grundpraktikum, Universität Würzburg, Modul C1, Versuch 49, Kennlinien von ¨ Vakuumdioden und Franck-Hertz-Versuch, 2021

[2] Meschede, Dieter, Gerthsen Physik, 25. Auflage, Springer-Verlag, Berlin, 2015

6 Appendix

differences between the main and secondary maxima of neon:

$$\Delta E_2^1 = (8.2 \pm 4.4)eV$$
$$\Delta E_3^2 = (3.6 \pm 8.1)eV$$
$$\Delta E_3^3 = (6.0 \pm 6.5)eV$$
$$\Delta E^{23} = (2.5 \pm 6.5)eV$$

superscript numbers specify the secondary maxima, subscript numbers the main maxima.

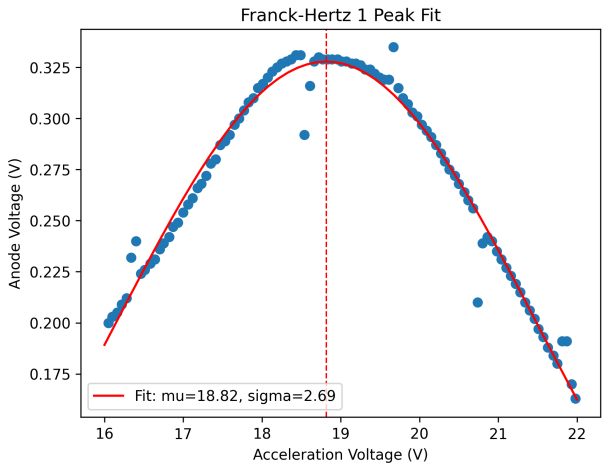

Fig. 5: Fit that was used to calculate the peak and the error for the first maximum of the neon Franck-Hertz line

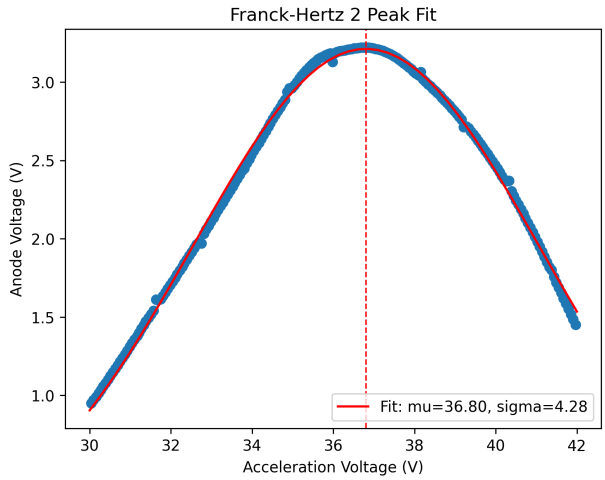

Fig. 6: Fit that was used to calculate the peak and the error for the second maximum of the neon Franck-Hertz line

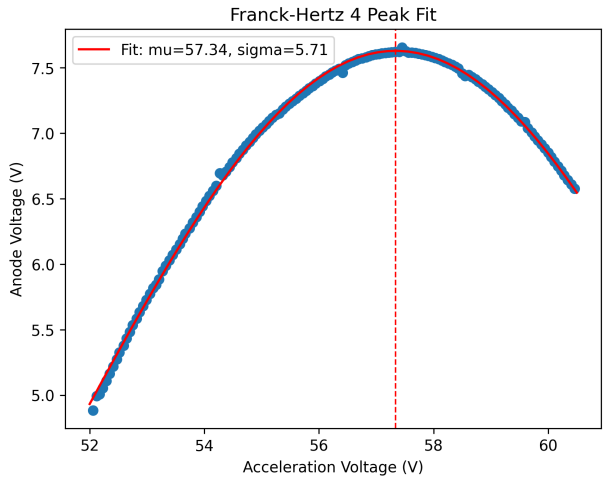

Fig. 7: Fit that was used to calculate the peak and the error for the third maximum of the neon Franck-Hertz line

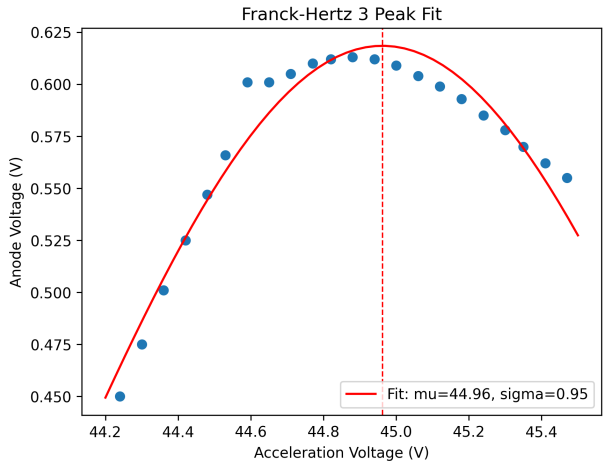

Fig. 8: Fit that was used to calculate the peak and the error for the first side maximum of the neon Franck-Hertz line

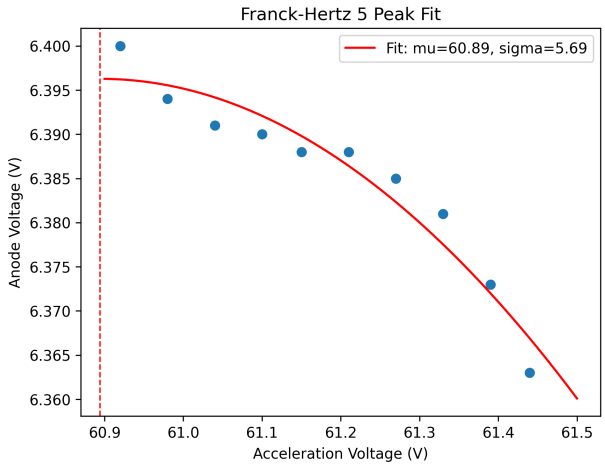

Fig. 9: Fit that was used to calculate the peak and the error for the second side maximum of the neon Franck-Hertz line

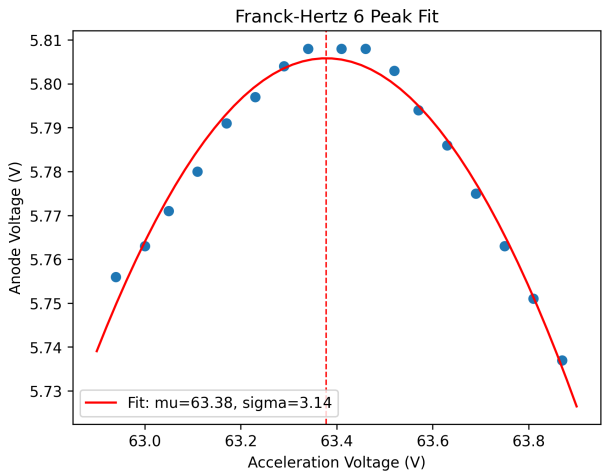

Fig. 10: Fit that was used to calculate the peak and the error for the third side maximum of the neon Franck-Hertz line

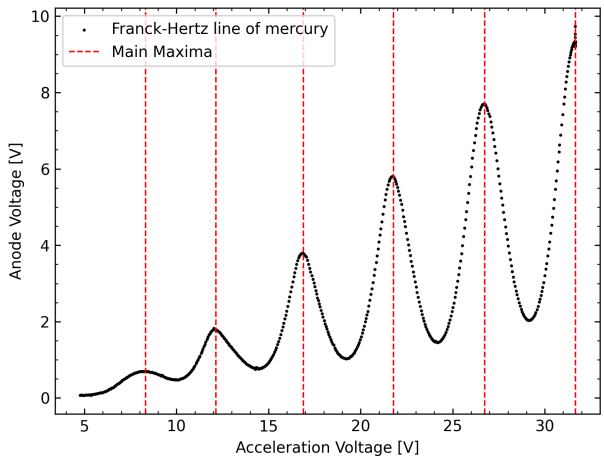

Fig. 11: plotted data of the Franck-Hertz line of mercury with maxima

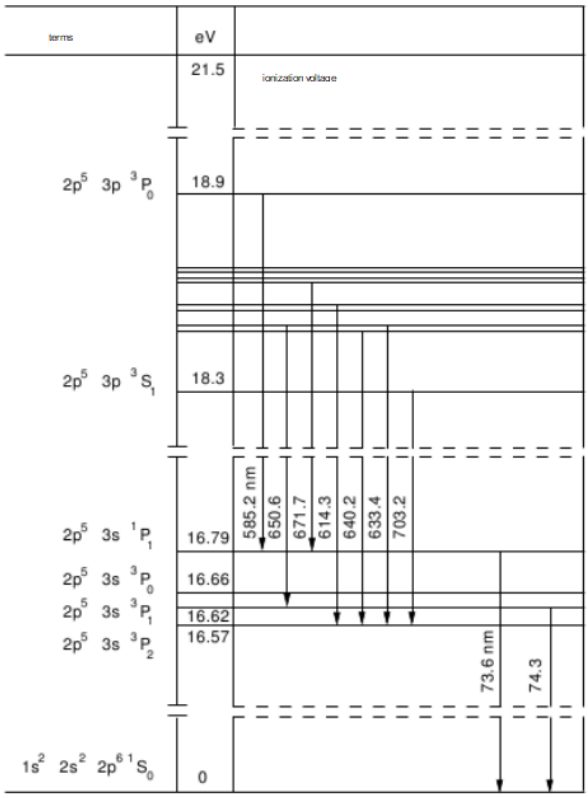

Fig. 12: term diagram of neon [1]

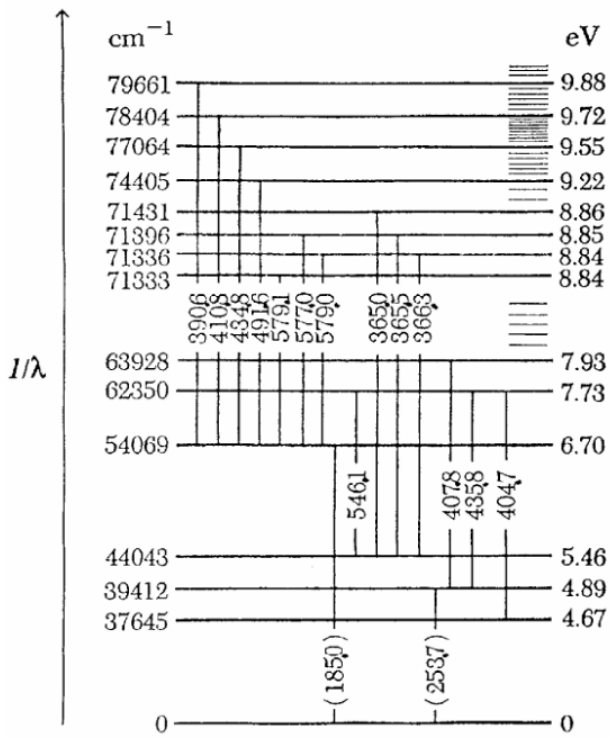

Fig. 13: term diagram of mercury [1]